

HealthConnect Clinic

Data Analytics Track — Week 5 Analytics Report

AnalystLab Africa Experience Lab | Prepared by: Moses Oluwatosin | Date: 2 September 2026

1. Week 4 Foundation — Brief Recap

This section is a brief transition, not a repeat of the Week 4 submission (docs/HealthConnect_Week4_Analysis_Document.docx).

- Problem defined: understand which factors are associated with appointment no-shows, to support HealthConnect's goal of reducing missed appointments.
- Week 4 approach: review data dictionary → inspect dataset → assess data quality → identify important variables → define business questions and KPIs.
- Resources used: HealthConnect_Appointment_Data.csv and HealthConnect_Data_Dictionary.xlsx. The Knowledge Base remains out of scope for this track.
- Key assumptions/limitations carried in: dataset is synthetic; findings are associative, not causal; missingness in distance/waiting time (<2%) still unaddressed; Sunday-appointment question still open.
- Week 5 goal (as proposed in Week 4): move from KPI identification to calculation, deepen EDA, and produce business insights and a visual report.

2. Data Preparation

Full data-quality assessment was completed in Week 4 (types, missingness, duplicates, consistency rules — all passed with zero violations). This week re-confirms that status is unchanged rather than repeating the assessment:

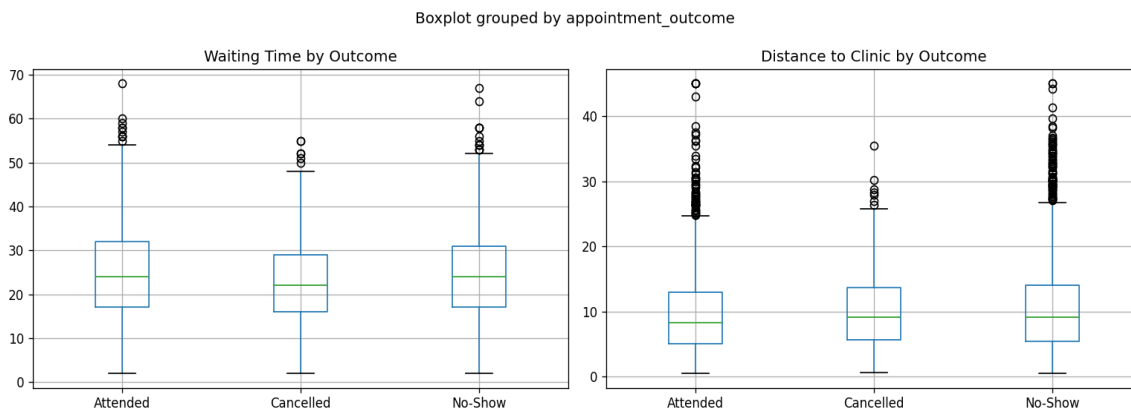
- Original dataset file remains untouched — no cleaning or transformation was required, matching the Week 4 verdict.
- Duplicate rows: 0. Duplicate appointment_id values: 0.
- Missingness unchanged: reminder_channel (27.3%, fully explained by reminder_sent = No), distance_to_clinic_km (1.8%), waiting_time_minutes (1.2%).
- All Week 4 consistency rules (lead-time recomputation, previous_no_shows ≤ previous_appointments, no negative values) re-checked and still hold.

No processed/derived dataset file was produced this week, since none was needed. Any derived columns (e.g. lead-time buckets) are created in-notebook for analysis only.

3. Exploratory Data Analysis (Deepened)

Week 4 identified booking_lead_days and previous_no_shows as strong signals and reminder_sent as moderate. This week extends the analysis into variables and relationships not yet fully explored.

3.1 Waiting Time & Distance



Medians for both `waiting_time_minutes` and `distance_to_clinic_km` stay close across outcome groups — consistent with Week 4's "weak signal" flag. Neither is a strong standalone predictor of no-shows in this dataset.

3.2 Cancellations, Analysed Separately from No-Shows

Cancellation rate stays roughly flat (~5%) across appointment types and across booking-lead-time buckets — unlike no-show rate, which climbs sharply with lead time. This supports treating Cancelled as a genuinely different operational event (planned, advance-notice non-attendance) rather than folding it into the no-show analysis, as assumed in Week 4.

3.3 Interaction: Lead Time × Prior No-Shows

Even at the shortest lead time (0-7 days), patients with 3+ prior no-shows show a materially higher no-show rate than first-time patients booked far in advance. This suggests prior history carries independent predictive value, not just as a proxy for lead time — relevant to KPI 5 and to segment targeting (Q5).

4. KPI Development

The 5 KPIs identified and linked to business questions in Week 4 are calculated here in full.

KPI 1 — No-Show Rate (Linked question: Q1)

Definition: $\text{No-Show appointments} \div \text{total scheduled appointments}$.

Why it matters: Primary metric for the clinic's core pain point — wasted capacity.

Calculated value: 48.5%

Interpretation: Nearly half of all scheduled appointments in this dataset are missed — a high baseline confirming the scale of the problem described in the business scenario.

KPI 2 — Attendance Rate (Linked question: Q1)

Definition: $\text{Attended} \div \text{total scheduled appointments}$.

Why it matters: Complements No-Show Rate for capacity-planning context.

Calculated value: 46.3%

Interpretation: Well under half of scheduled slots are actually used by the booked patient — the practical capacity-utilisation problem HealthConnect needs to solve.

KPI 3 — Cancellation Rate (Linked question: Q1, Q3)

Definition: $\text{Cancelled} \div \text{total scheduled appointments}$.

Why it matters: Separates planned non-attendance (advance notice, slot reallocatable) from unplanned no-shows.

Calculated value: 5.3%

Interpretation: Cancellations are a small share of non-attendance overall — the larger problem is unplanned no-shows, where intervention effort should concentrate.

KPI 4 — Reminder Effectiveness (Linked question: Q4)

Definition: No-show rate when a reminder was sent vs. not, split by channel.

Why it matters: Measures whether the reminder programme — and which channel — is worth the operational cost of running.

Calculated value: 47.4% (sent) vs 51.4% (not sent) — a ~4-point reduction. SMS is the best-performing channel among those tested.

Interpretation: The effect is real but modest. Reminders are a useful, low-cost lever, but not sufficient alone to solve a 48.5% no-show rate.

KPI 5 — Repeat No-Show Rate (Linked question: Q2, Q5)

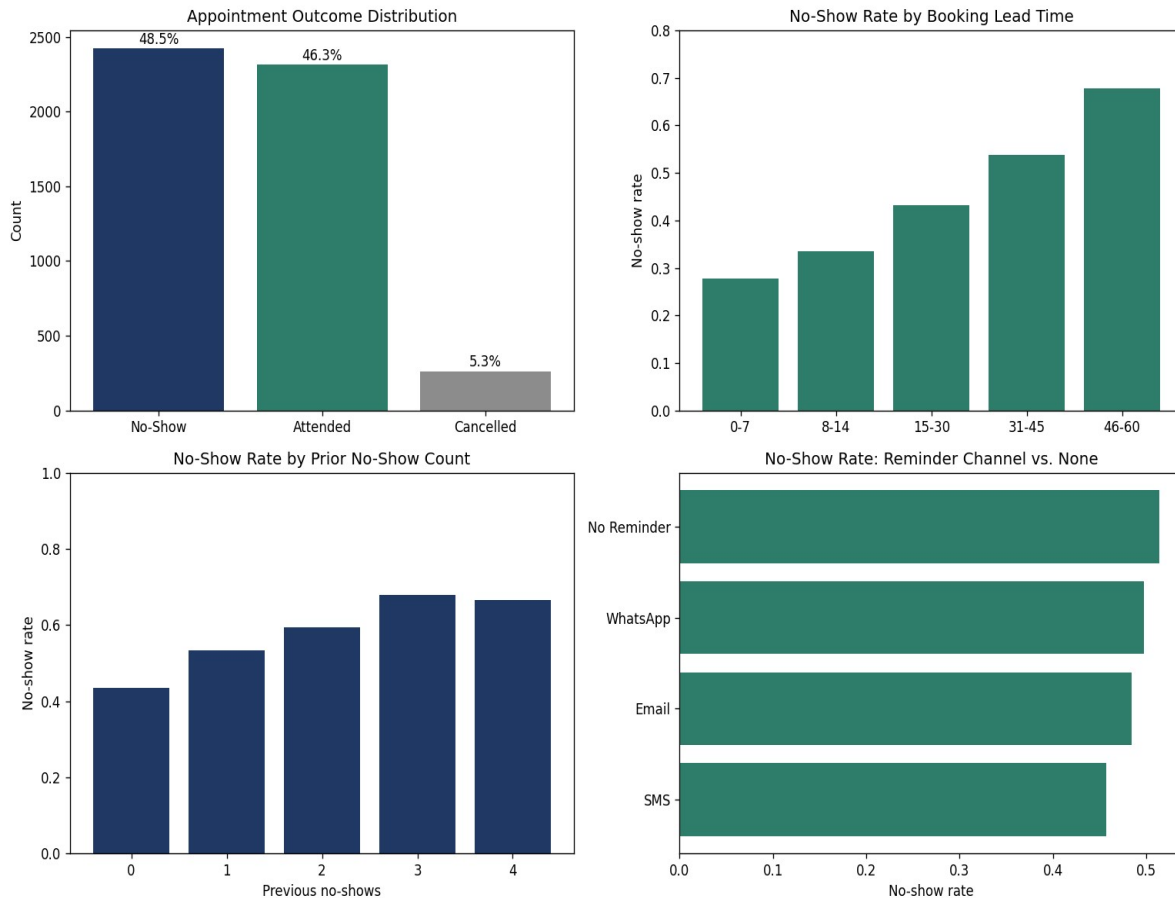
Definition: Share of appointments from patients with ≥ 2 prior no-shows, and the no-show rate within that segment.

Why it matters: Identifies a concentrated high-risk group for targeted intervention rather than treating all patients identically.

Calculated value: 10.6% of appointments come from patients with 2+ prior no-shows; within that segment, the no-show rate is 61.0% — well above the 48.5% overall baseline.

Interpretation: Prior history is a strong, actionable predictor — a natural target for a stricter reminder protocol or confirmation call, cheaper than intervening on the full patient base.

5. Visual Report / Analytical Dashboard



6. Business Insights & Recommendations

- Nearly half of appointments are missed (48.5% no-show) — the problem is significant enough to justify active intervention, not just monitoring.
- Booking lead time is the clearest driver found (28% → 68% no-show rate as lead time grows). Consider tighter booking windows for high-demand slots, or stronger reminder protocols for appointments booked far in advance.
- Prior no-show history is a strong, largely independent risk signal. A "repeat no-show" flag could trigger a different workflow — confirmation call, deposit, or planned overbooking for that slot.
- Reminders help but only modestly (~4-point reduction); SMS outperforms other channels. Reminders are a supporting lever, not the primary fix — but channel choice is a low-cost, easy win.
- Cancellations are small (~5%) and stable, behaving differently from no-shows. Cancellation handling (rebooking the freed slot) is a simpler, separate process and shouldn't be conflated with no-show intervention design.
- Waiting time and distance to clinic are not strong standalone predictors in this dataset — effort is better spent on the levers above based on current evidence.

7. Cross-Track Collaboration

Track interacted with: Data Science (conceptual handoff).

Information/output exchanged: The variable-signal findings from this report — booking_lead_days and previous_no_shows as strong predictors, reminder_sent/reminder_channel as moderate, distance_to_clinic_km/waiting_time_minutes as weak — form a natural candidate-feature shortlist for a no-show prediction model.

Why relevant: The Data Science track's own Week 4 output proposed a similar feature set from first principles; this report supplies the quantified evidence (actual rates, not just hypothesis) behind that choice.

What changed/improved as a result: The Data Science track can prioritise booking_lead_days, previous_no_shows, and reminder_sent/reminder_channel as first-pass model features, and deprioritise distance/waiting time unless a later interaction effect is found.

8. Assumptions, Limitations, Risks & Dependencies — Week 5 Update

Item	Week 4 status	Week 5 status
Dataset is synthetic; findings illustrative	Assumption	Unchanged
Cancelled treated as distinct from No-Show	Assumption	Confirmed — cancellations behave differently (flat rate vs. lead time)
Missing data in distance/waiting time (<2%)	Limitation	Unchanged / still open — lower priority since weak signal anyway
Sunday appointment records need stakeholder confirmation	Risk	Unresolved — carried forward
Findings are associative, not causal	Risk	Still the top risk — no significance testing performed yet
Interaction effects underexplored	— (new)	New this week — only one interaction tested (lead time × prior no-shows); others untested

9. Limitations & Recommendations for Week 6

- All findings remain descriptive/associative — statistical significance testing (chi-square, logistic regression) is needed before treating any variable as a confirmed driver.
- Only one variable interaction was tested; other combinations (e.g. reminder channel × prior history) remain unexplored.
- The Sunday-appointment and missing-data open items from Week 4 were not addressed this week and are carried forward as-is.
- Week 6 focus: significance testing on the KPIs above, broader interaction analysis, and translating the 5 insights into a stakeholder-ready recommendation with estimated impact.

Full code: [notebooks/HealthConnect_Week5_Analytics.ipynb](#) | Summary: [docs/HealthConnect_Week5_Project_Summary.docx](#) | Navigation: [README.md](#)